Appendix

Additional Aspects

Training Time

Training times vary substantially across approaches, as shown in Tab. 1. The baselines and foundation models Chronos and TabPFN require no task-specific training. Statistical models, as well as Ridge and Ridge_M, train in less than one second. N-BEATS requires 25 seconds, whereas TFT takes approximately 1.5 minutes. Trace-level PPM approaches are computationally more expensive, with training times of approximately 18–23 minutes per log. SIMOD takes the longest with more than 4 hours. This suggests that optimizing predictions at the trace level requires substantially greater training effort without translating into improved process-level forecasting accuracy.

Table 1: Training times per approach, averaged across logs. **–** indicates no training phase.

Baseline			Statistical				ML				FM		SIM		PPM		
Naïve	Val. Avg.		Proph.	ETS	SARIM.	STL	Ridge	Ridge_M	N-BEA.	TFT	Chronos	TabPFN	PMSD	Simod	GLSTM	PGT	PT
–	–		<1 s	<1 s	1 s	<1 s	<1 s	<1 s	25 s	1 m 28 s	–	–	1 s	4 h 27 m	22 m 43 s	17 m 47 s	20 m 45 s

Effect of Inter-Case Features in Process-Level Predictions with Trace-Level Approaches

A disadvantage of trace-level PPM approaches is that they do not account for interactions between different traces, called inter-case effects, as they operate on individual traces by design. To investigate whether the incorporation of such inter-case features helps to better perform process-level predictions, we extend GLSTM and PT_RT with the inter-case features proposed in Gunnarsson et al. (2024). Besides these additional features, training settings remain unchanged. To ensure meaningful inter-case features in GLSTM's suffix prediction, we predict the next event of all running instances, choose the soonest, and update the state of the inter-case features, thus iterating over the full set of active traces and not trace-by-trace.

Tab. 2 shows the relative change in CTP and TTP in the inter-case setting compared to the default setting without inter-case features. Inter-case features do not systematically improve process-level prediction accuracy. Across both models, the majority of cells are positive, indicating *higher* MAE compared to the variant without inter-case features. Improvements, where they occur, are inconsistent across prefix configurations and do not generalize across datasets.

The effect differs between the two architectures and is more favorable for GLSTM. For PT_RT, inter-case features degrade performance almost universally. The MAE change even surpass $+300\%$ for O2C Flat across all configurations. This is likely since the remaining time prediction is done only once per trace: for *Plain* and *First* at the start of each trace, for *Half* at the middle. Inter-case features thus only depend on a single prefix snapshot, making them uninformative and rather disturbing the prediction model. For GLSTM, improvements are sometimes visible but not consistent. On the synthetic logs, inter-case features help for some LoAp scenarios—e.g., Seasonal (-14 to -35%), Drift (-10 to -20%)—but degrade others, including most O2C

Table 2: Relative MAE-change (%) of inter-case models compared to default versions. Negative values indicate improvement.

			CTP						*TTP*					
			GLSTM			PT_{RT}			GLSTM			PT_{RT}		
			Plain	First	Half	Plain	First	Half	Plain	First	Half	Plain	First	Half
Synthetic	Loan	Flat	−27.91	−12.04	26.72	61.11	111.58	73.37	−14.06	−11.11	19.08	129.60	122.41	56.97
		Seasonal	−35.70	−20.08	−20.54	26.81	24.54	2.99	−34.10	−20.37	−13.67	21.79	21.00	3.31
		Trend	−7.25	8.82	58.25	161.77	246.02	57.12	−4.07	11.11	57.49	262.77	267.82	46.89
		Drift	−18.12	−11.58	−19.65	−24.20	−27.84	−8.64	−15.87	−10.34	−9.98	−24.25	−24.83	−6.08
		Combined	34.48	34.05	25.11	69.57	58.02	51.09	31.19	47.87	25.99	71.40	77.35	54.92
	O2C	Flat	46.69	−5.99	148.69	334.11	219.53	483.07	−28.21	19.17	301.90	321.02	323.93	357.48
		Seasonal	−9.16	−56.16	76.38	0.00	11.36	9.22	−18.03	−31.62	81.84	−1.47	10.84	11.55
		Trend	−3.53	8.71	71.54	4.93	17.93	33.00	−14.00	10.46	113.35	12.87	11.64	42.70
		Drift	9.15	−62.32	4.80	7.43	18.11	17.44	24.60	6.89	29.82	19.30	19.20	20.42
		Combined	−1.31	−22.11	42.16	28.09	98.14	179.89	−4.73	−25.08	58.73	165.26	164.79	382.02
Real-life		bpi12-a	56.42	75.64	44.47	163.47	192.67	333.45	20.87	89.83	27.75	111.15	94.94	224.45
		bpi15-1	−18.25	−80.69	−26.32	145.55	54.01	70.86	−2.86	−13.68	−4.28	−17.73	−5.83	−14.06
		bpi15-2	1.19	−52.90	1.88	11.91	1.23	40.94	−12.24	−1.08	6.31	−0.64	0.11	−1.42
		bpi17-o	120.55	48.62	359.99	0.77	−12.15	10.11	70.96	36.97	98.12	−5.00	−9.98	13.01
		bpi20-d	−6.84	−10.93	80.72	37.20	27.03	163.13	−6.40	−5.59	5.20	6.70	1.85	4.33
		bpi20-i	53.14	25.10	3.26	−16.56	−24.61	146.37	27.60	22.47	−1.71	−22.74	−25.68	45.37
		helpdesk	10.70	93.58	−31.91	9.29	−2.49	44.32	121.55	37.64	−40.25	18.66	13.11	0.22
		sepsis	−38.02	−39.72	−41.99	−0.91	−1.99	2.90	−0.92	−0.80	−3.77	0.77	0.41	1.30

settings. In the real-life logs, the pattern is similarly inconsistent. GLSTM with inter-case features improves substantially on bpi15-1 (up to -81% for CTP) and sepsis (-38 to -42% for CTP), but degrades on bpi17-o ($+49$ to $+360$% for CTP) and bpi12-a ($+44$ to $+76$% for CTP). Even within the same log, the direction can flip across prefix configurations: on bpi20-d, *Plain* and *First* improve (-7 to -11%) while *Half* degrades ($+81$%).-

Critically, despite occasional improvements over the default trace-level variant, the inter-case models almost never are competitive with the process-level approaches. Across all 216 model–dataset–series combinations, only a single inter-case MAE reaches the top three among all approaches: GLSTM *First* on bpi15-1 for CTP, which reaches the second best MAE of all approaches investigated. This indicates that the problem of trace-level PPM not accounting for interactions between traces is not solved by adding inter-case features.

Sensitivity to Training Data Size

We further assess the sensitivity of selected process-level and trace-level approaches to the amount of available training data. To this end, we progressively reduce the training portion of the LoAp logs while correspondingly increasing the test set, resulting in four training-set sizes: 70% (original), 60%, 40%, and 20%. We evaluate five approaches under these configurations and report the results for CTP and TTP in Figure 1.

Three main observations emerge. First, the trace-level approach PT_{su} yields the highest MAE across all training-set sizes for both CTP and TTP. Second, for CTP, PT_{su} shows the smallest relative performance degradation as the training set is reduced. However, this apparent robustness is largely due to its already poor performance with larger training sets, leaving comparatively little room for further deterioration. Third, none of the evaluated approaches exhibits a consistently strong sensitivity or robustness to training-set size across both prediction

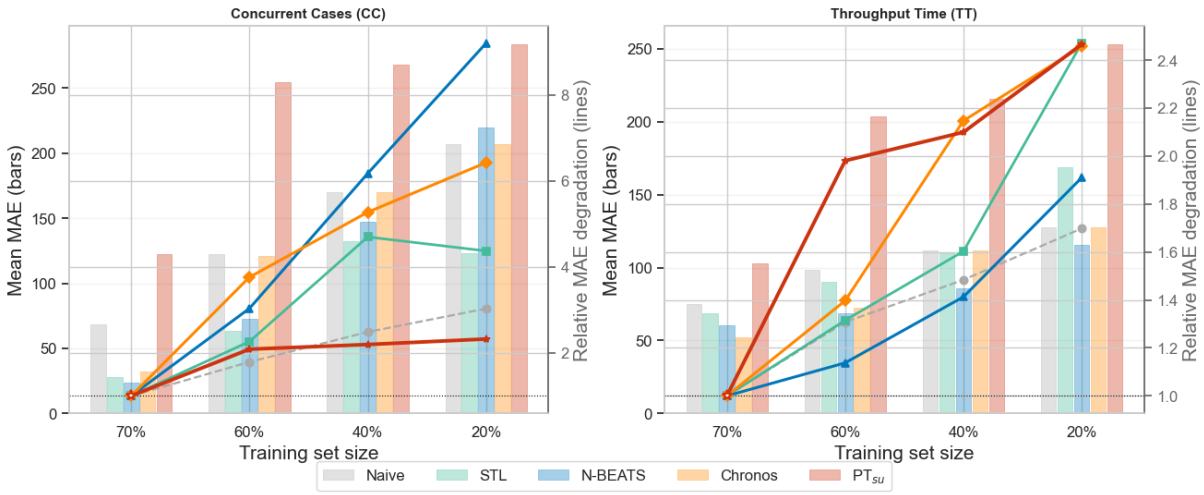


Figure 1: Sensitivity of different approaches to training data size, averaged over all synthetic LoAp logs. Bars show absolute MAE per approach for varying training set sizes. Lines show the relative degradation of the approaches w.r.t. their original performance when the full training set (70%) is given.

tasks. Overall, reducing the amount of training data affects performance, but no clear and consistent pattern emerges across approaches and tasks.

Hyperparameter Selection

Table 3: Hyperparameter search spaces for all tunable models. All time series models are selected via exhaustive grid search on the validation set; the trace-level models (ProcessTransformer, GenerativeLSTM) use Bayesian optimization. Fixed parameters not listed here are held constant across all candidates.

Model	Hyperparameter	Search Space
ETS	Error component	{add, mul}
	Trend component	{none, add, mul}
	Seasonal component	{none, add, mul}
	Damped trend	{True, False}
SARIMAX	ARIMA order (p, d, q)	{(1,0,0),(1,1,0),(0,1,1),(1,1,1)}
	Seasonal order (P, D, Q, s)	{(0,0,0,0),(1,0,0,7),(0,1,1,7),(1,1,1,7)}
STL	Loess period	$\{7, 14\}$
	Error component	{add, mul}
	Trend component	{none, add}
Ridge / Ridge$_{\text{M}}$	Lags (lookback)	$\{7, 14, 28\}$
	Regularisation α	$\{0.01, 0.1, 1, 10\}$
N-BEATS	Input chunk length	$\{14, 28\}$
	Output chunk length	$\{7, 14\}$
	Layer width	$\{64, 128\}$
Prophet	Changepoint prior scale	$\{0.01, 0.05, 0.5\}$
	Seasonality prior scale	$\{1, 10\}$
	Seasonality mode	{add, mul}
TFT	Input chunk length	$\{14, 28\}$
	Output chunk length	$\{7, 14\}$
	Hidden size	$\{16, 32\}$
TabPFN	Lookback n_{steps}	$\{7, 14, 28\}$
PT	Attention heads	$\{2, 4\}$
	Batch size	$\{16, 32\}$
	Epochs	$\{20, 50\}$
	Learning rate	$\{5\times10^{-4}, 10^{-3}, 5\times10^{-3}\}$
GLSTM	Model type	{shared_cat, concatenated}
	LSTM activation	{selu, tanh}
	Learning rate	$\{5\times10^{-4}, 10^{-3}, 2\times10^{-3}\}$
	Dropout	$\{0.1, 0.2, 0.3\}$

Robustness

We here report the average MAE ± standard deviation in CTP across three random seeds for all the models, indicating their robustness to random weight initialization. We show only CTP in the synthetic logs, the rest can be found in the file `robustness_by_dataset.xlsx` in the `/results` folder. Note that many approaches are deterministic whereas some ML-based TS models show fluctuations and trace-level PPM approaches vary even more.

Table 4: Average MAE ± standard deviation in CTP across three random seeds in LoAp logs.

			LoAp (simple)				
			Flat	Season.	Trend	Drift	Comb.
Basel.		Naïve	14.13 ± 0.00	15.38 ± 0.00	45.93 ± 0.00	15.71 ± 0.00	187.47 ± 0.00
		Val. Avg.	27.71 ± 0.00	29.65 ± 0.00	109.32 ± 0.00	23.99 ± 0.00	254.37 ± 0.00
Statistical		ETS	14.13 ± 0.00	10.97 ± 0.00	345.35 ± 0.00	49.73 ± 0.00	247.01 ± 0.00
		SARIMAX	75.81 ± 0.00	101.08 ± 0.00	44.94 ± 0.00	99.24 ± 0.00	12.36 ± 0.00
		STL	14.12 ± 0.00	13.75 ± 0.00	48.28 ± 0.00	12.96 ± 0.00	45.30 ± 0.00
		Prophet	645.10 ± 0.00	561.89 ± 0.00	121.06 ± 0.00	173.59 ± 0.00	23.37 ± 0.00
ML		Ridge	12.43 ± 0.00	15.26 ± 0.00	17.68 ± 0.00	23.99 ± 0.00	85.28 ± 0.00
		Ridge_{M}	18.50 ± 0.00	21.87 ± 0.00	19.62 ± 0.00	23.49 ± 0.00	96.44 ± 0.00
		N-BEATS	20.14 ± 2.22	19.26 ± 0.54	42.03 ± 13.68	28.51 ± 5.60	110.41 ± 88.58
		TFT	29.05 ± 1.86	32.91 ± 2.27	180.58 ± 84.44	52.07 ± 6.92	292.52 ± 141.89
FM		Chronos	8.93 ± 2.59	12.52 ± 2.10	55.51 ± 4.12	12.12 ± 1.50	64.10 ± 6.12
		TabPFN	21.25 ± 3.10	17.43 ± 4.12	139.12 ± 5.10	33.19 ± 1.67	167.98 ± 5.23
SIM		PMSD	19.75 ± 0.00	12.83 ± 0.00	62.91 ± 0.00	22.08 ± 0.00	79.70 ± 0.00
		Simod	680.33 ± 0.00	632.28 ± 0.00	244.40 ± 0.00	615.15 ± 0.00	419.55 ± 0.00
		AgentSimulator	150.35 ± 0.00	100.89 ± 0.00	181.67 ± 0.00	236.58 ± 0.00	515.89 ± 0.00
Trace-Level PPM	Plain	GLSTM	162.46 ± 37.95	243.76 ± 18.57	205.67 ± 39.94	158.27 ± 20.93	615.29 ± 50.34
		PGT	133.21 ± 2.65	91.30 ± 1.81	167.18 ± 1.56	88.49 ± 2.25	361.27 ± 3.40
		PT_{RT}	106.10 ± 19.34	168.92 ± 0.49	162.25 ± 44.02	149.34 ± 13.10	568.06 ± 146.47
		PT_{su}	74.17 ± 19.71	116.99 ± 13.05	76.47 ± 41.57	76.34 ± 8.85	413.50 ± 102.49
	First	GLSTM	224.71 ± 29.78	240.72 ± 30.61	301.21 ± 30.34	167.39 ± 15.22	659.89 ± 48.05
		PF Amiri	62.68 ± 2.65	62.23 ± 2.00	91.33 ± 1.54	38.38 ± 2.25	430.12 ± 3.40
		PT_{su}	105.91 ± 60.22	130.76 ± 12.84	181.27 ± 45.78	90.64 ± 7.67	467.94 ± 94.19
		PT_{RT}	158.77 ± 55.91	182.06 ± 0.50	228.01 ± 104.41	162.86 ± 12.90	617.57 ± 134.33
	Half	GLSTM	121.97 ± 19.89	127.22 ± 13.18	165.02 ± 16.98	89.00 ± 11.23	383.05 ± 36.37
		Amiri	112.82 ± 8.72	108.38 ± 6.20	181.87 ± 36.60	89.98 ± 9.03	357.04 ± 7.06
		PT_{su}	114.72 ± 42.72	92.13 ± 6.30	137.14 ± 23.64	114.30 ± 3.42	78.10 ± 41.27
		PT_{RT}	104.50 ± 27.63	125.67 ± 5.81	155.36 ± 48.30	92.95 ± 2.45	381.65 ± 88.13

Table 5: Average MAE ± standard deviation in CTP across three random seeds in O2C logs.

			O2C			
		Flat	Season.	Trend	Drift	Comb.
Basel.	Naïve	26.43 ± 0.00	80.89 ± 0.00	485.04 ± 0.00	48.53 ± 0.00	1021.45 ± 0.00
	Val. Avg.	22.55 ± 0.00	48.64 ± 0.00	596.61 ± 0.00	47.06 ± 0.00	1337.14 ± 0.00
Statistical	ETS	21.31 ± 0.00	40.69 ± 0.00	171.14 ± 0.00	32.94 ± 0.00	172.44 ± 0.00
	SARIMAX	73.03 ± 0.00	41.21 ± 0.00	71.62 ± 0.00	67.69 ± 0.00	105.79 ± 0.00
	STL	24.76 ± 0.00	22.26 ± 0.00	159.55 ± 0.00	32.59 ± 0.00	100.02 ± 0.00
	Prophet	1498.57 ± 0.00	1536.06 ± 0.00	1747.54 ± 0.00	395.31 ± 0.00	637.37 ± 0.00
ML	Ridge	67.84 ± 0.00	124.77 ± 0.00	114.98 ± 0.00	128.07 ± 0.00	236.02 ± 0.00
	Ridge_{M}	53.89 ± 0.00	54.68 ± 0.00	1100.86 ± 0.00	22.85 ± 0.00	1334.02 ± 0.00
	N-BEATS	41.57 ± 7.14	58.46 ± 4.27	688.26 ± 271.25	42.58 ± 1.91	236.24 ± 103.67
	TFT	101.00 ± 47.03	108.87 ± 58.36	847.55 ± 306.49	107.82 ± 39.20	1436.94 ± 709.97
FM	Chronos	45.64 ± 5.14	147.12 ± 7.20	502.24 ± 10.10	65.84 ± 3.65	274.44 ± 7.14
	TabPFN	67.61 ± 2.14	65.14 ± 4.14	1,312.05 ± 50.78	68.60 ± 5.20	1,124.43 ± 100.14
SIM	PMSD	47.24 ± 0.00	70.16 ± 0.00	263.24 ± 0.00	32.72 ± 0.00	70.97 ± 0.00
	Simod	993.10 ± 0.00	1,415.46 ± 0.00	2,090.27 ± 0.00	1,096.04 ± 0.00	3,618.08 ± 0.00
	AgentSimulator	744.40 ± 0.00	963.51 ± 0.00	701.74 ± 0.00	1,347.46 ± 0.00	2,400.08 ± 0.00
Trace-Level PPM – Plain	GLSTM	1194.79 ± 43.59	1185.21 ± 61.95	1457.74 ± 156.73	1139.23 ± 34.41	3056.05 ± 58.62
	PGT	466.13 ± 10.64	1069.96 ± 19.21	1958.61 ± 17.02	1052.34 ± 19.42	3262.59 ± 61.64
	PT_{RT}	776.67 ± 439.73	1574.14 ± 15.80	2315.54 ± 14.07	1445.93 ± 31.68	3604.15 ± 398.87
	PT_{su}	603.40 ± 194.59	1208.09 ± 56.40	1981.31 ± 33.65	1157.70 ± 62.86	2959.47 ± 223.89
Trace-Level PPM – First	GLSTM	1032.27 ± 44.90	1553.39 ± 99.20	2318.13 ± 48.69	1466.30 ± 22.09	3793.55 ± 22.75
	PGT	405.28 ± 10.64	398.52 ± 19.21	432.82 ± 17.02	393.45 ± 19.42	1792.29 ± 61.64
	PT_{su} full	782.70 ± 171.69	850.94 ± 88.36	1204.94 ± 75.61	831.86 ± 97.43	2053.87 ± 422.49
	PT_{RT} full	935.45 ± 452.93	1257.72 ± 28.01	1542.26 ± 38.72	1087.59 ± 62.23	2713.15 ± 749.25
Trace-Level PPM – Half	GLSTM	746.22 ± 44.03	688.11 ± 67.67	799.05 ± 129.59	667.06 ± 25.08	2225.76 ± 86.94
	Amiri	454.02 ± 32.89	446.84 ± 160.43	656.11 ± 65.18	489.40 ± 82.85	1604.85 ± 261.43
	PT_{su}	214.18 ± 111.22	270.64 ± 67.12	362.62 ± 67.58	209.48 ± 76.68	1021.73 ± 301.95
	PT_{RT}	584.76 ± 384.99	837.21 ± 17.13	1009.88 ± 39.91	694.59 ± 1.77	1910.66 ± 736.50

References

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